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Escuela de Ciencias Aplicadas e Ingeniería

Lecture 10

GAN and CGAN

Agenda

➤ GAN

➤ cGAN

➤ Example



Input



Monet



Van Gogh



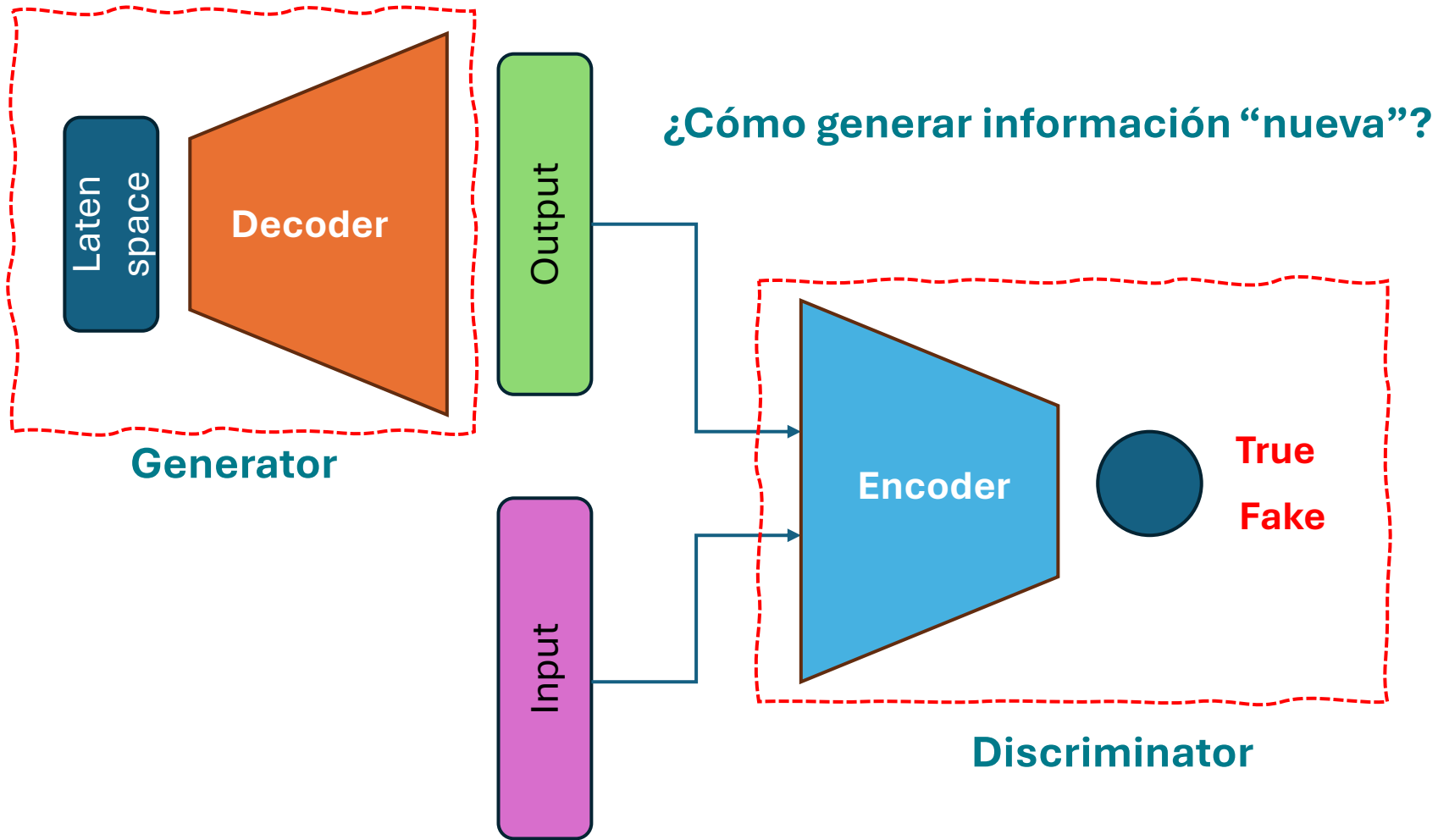
Cezanne



Ukiyo-e

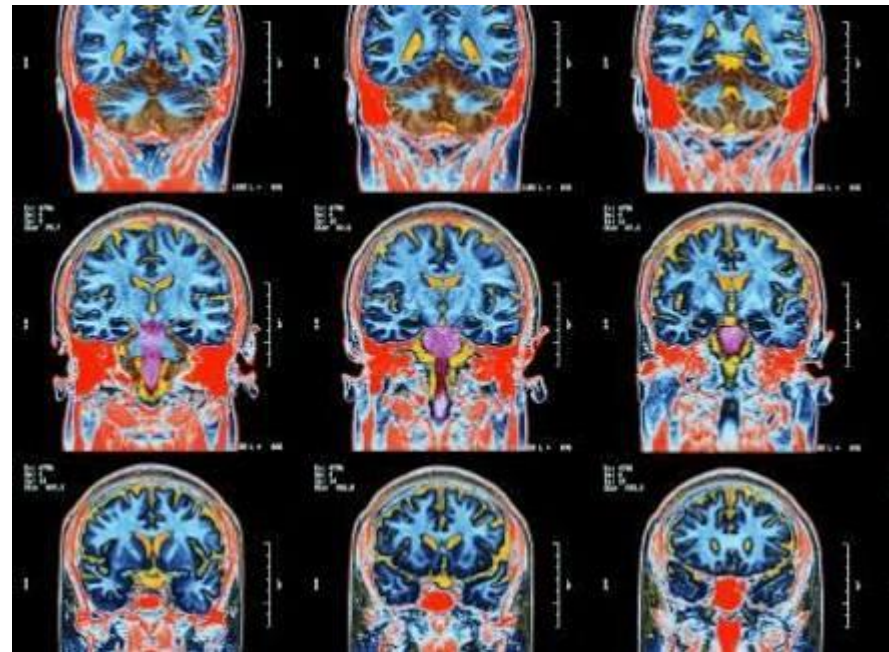
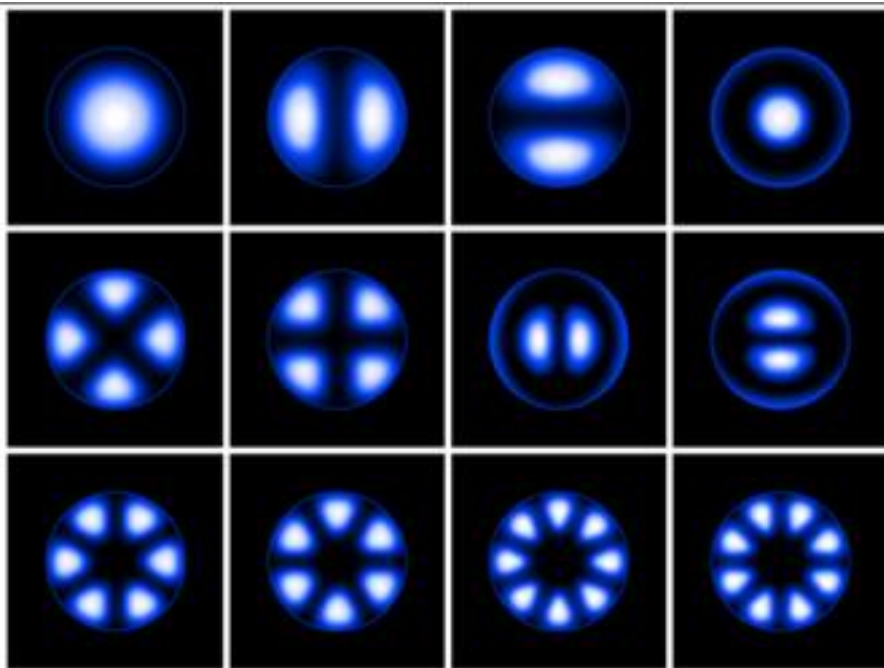


Modelos -GAN



Modelos -GAN

Los datos presentan una característica fundamental: **siguen una distribución estadística determinada**. El objetivo de un modelo generativo, como una **GAN**, es aprender dicha distribución a partir de los datos disponibles y **reproducirla con el mayor grado de fidelidad posible**.



$$x \sim P_{Data}(x)$$

$$GAN \rightarrow P_G(x)$$

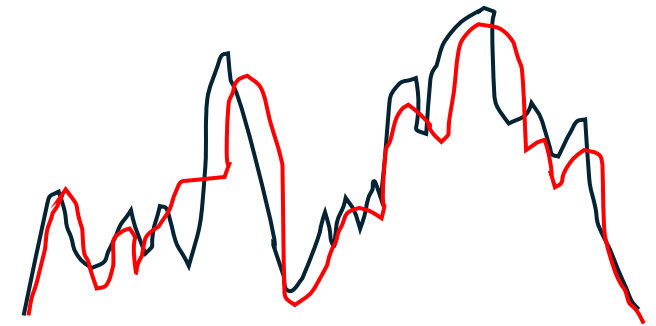
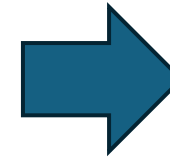
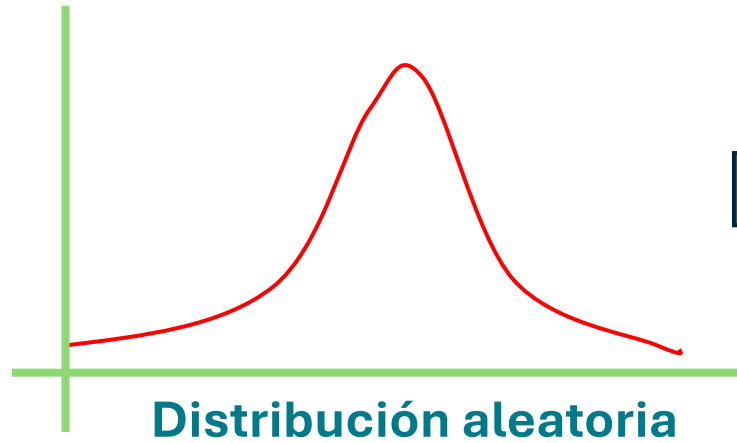
$$P_G(x) \cong P_{Data}(x)$$

Modelos -GAN

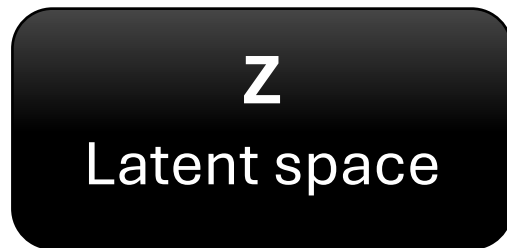
La idea es relativamente simple: a partir de datos aleatorios provenientes de una distribución sencilla, generar datos que sigan una distribución particular y más compleja.

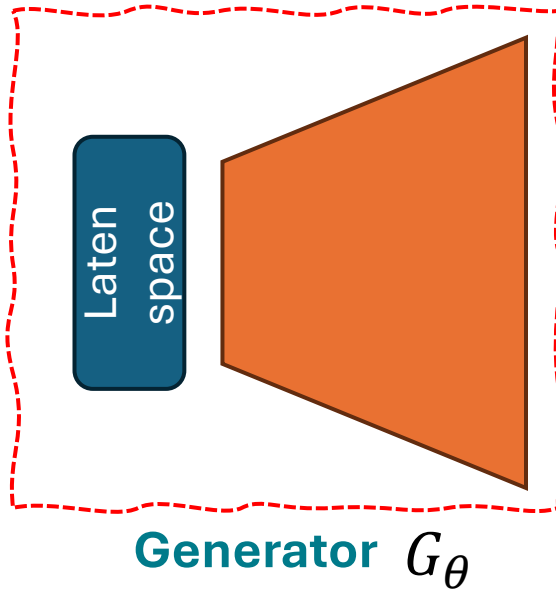


$$x \sim P_{Data}(x)$$



$$P_G(x) \cong P_{Data}(x)$$





¿Qué es el latent space z ? El **espacio latente** es un espacio de menor dimensión del cual se extraen vectores aleatorios z . Estos vectores actúan como entrada del generador y contienen la información a partir de la cual la red construye nuevas muestras.

$$z \sim p_z(z)$$

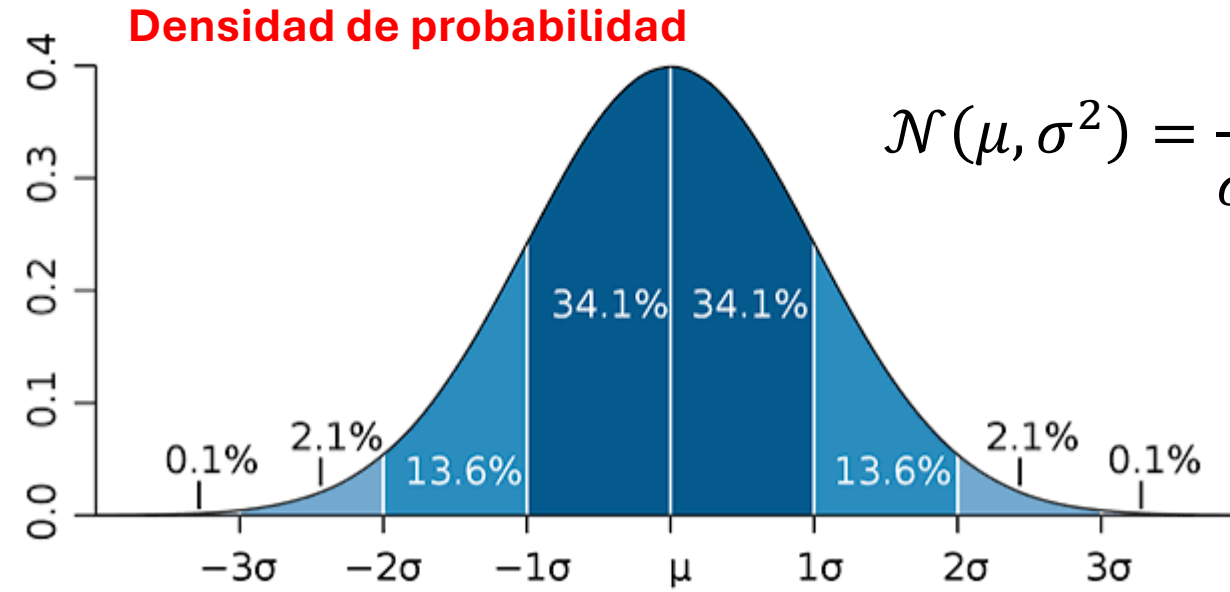
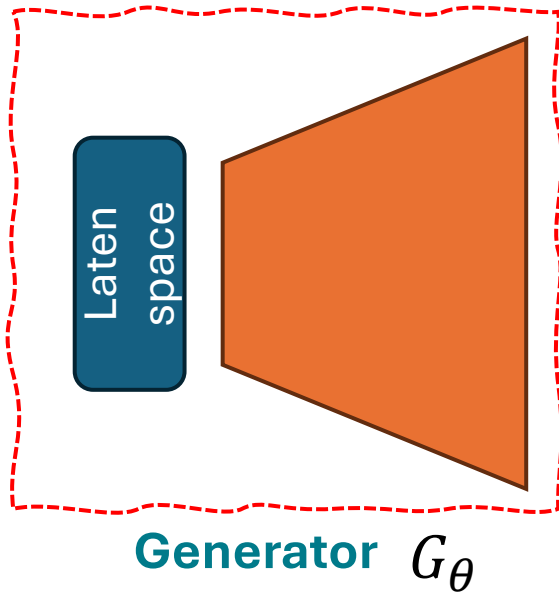
$$p_z(z) = \mathcal{N}(0, I)$$

$$z \xrightarrow{G_\theta} \hat{x}$$

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \xrightarrow{\quad} \begin{matrix} z \sim \mathcal{N}(0, I) \\ \mathcal{N}(\mu, \sigma^2) \\ p_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \end{matrix}$$

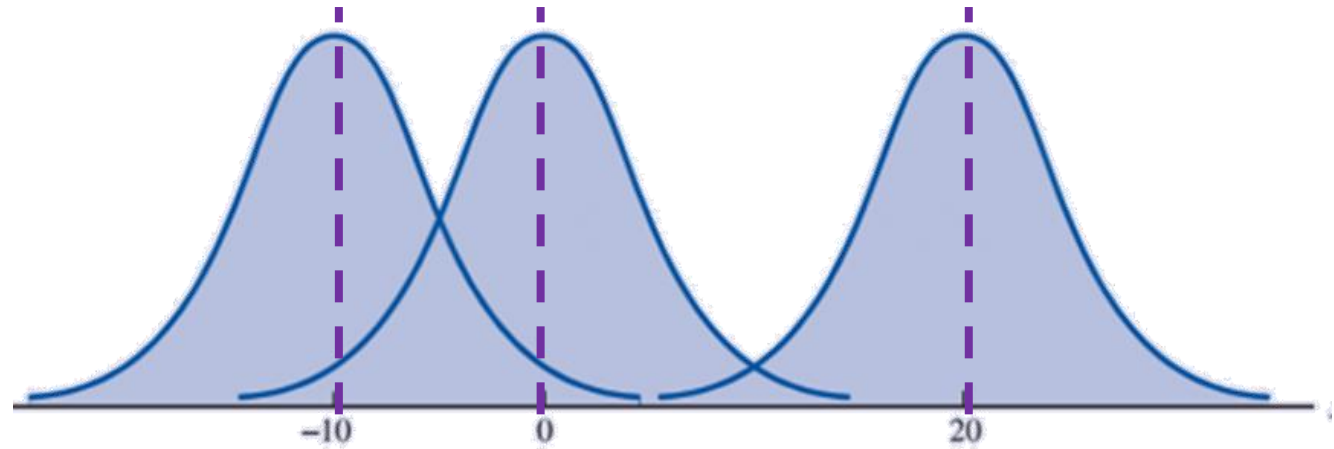
$$\xrightarrow{\quad} \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

Modelos -GAN

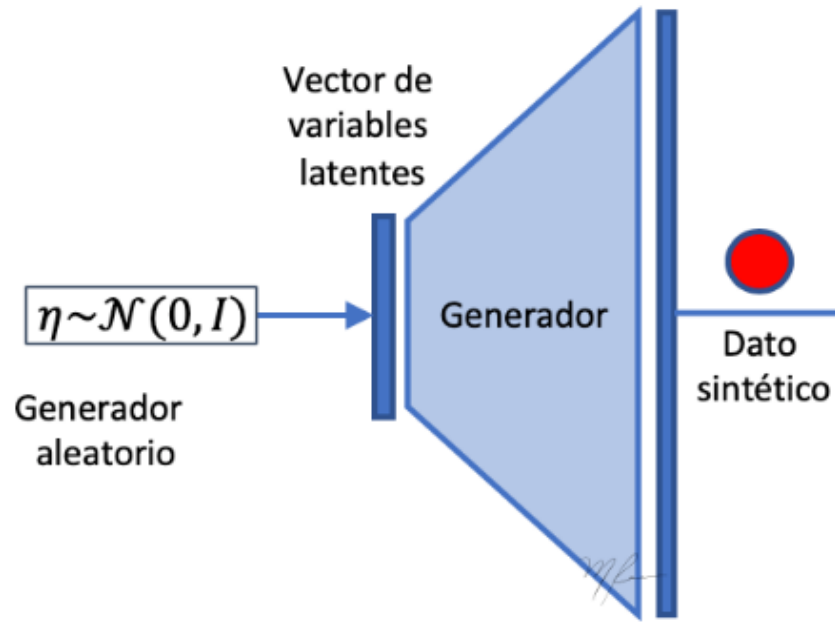


$$\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \rightarrow \begin{matrix} z \sim \mathcal{N}(0, I) \\ \mathcal{N}(\mu, \sigma^2) \\ p_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \end{matrix}$$



Modelos -GAN



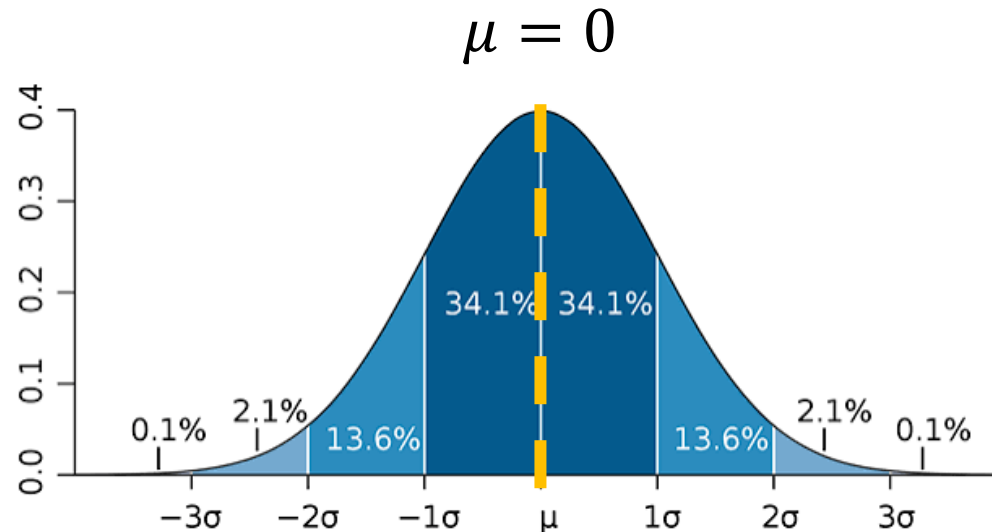
$$z \sim \mathcal{N}(0, I)$$

$\mu = 0$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & cov(z_1, z_2) & \dots & cov(z_1, z_n) \\ cov(z_2, z_1) & \sigma_2^2 & \dots & cov(z_2, z_n) \\ \vdots & \vdots & \ddots & \vdots \\ cov(z_n, z_1) & cov(z_n, z_2) & \dots & \sigma_n^2 \end{bmatrix}$$

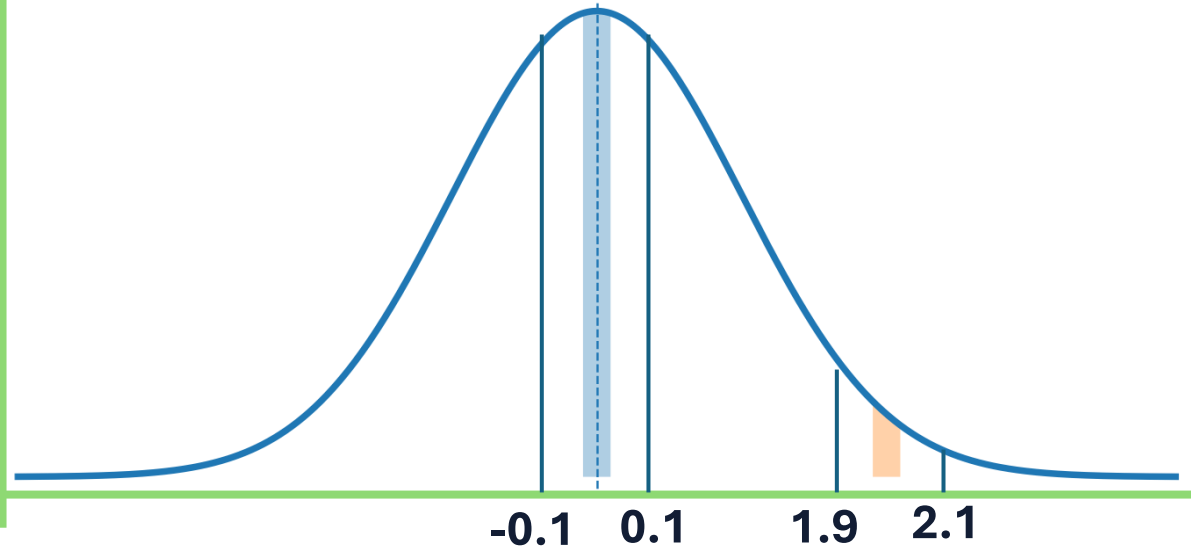
$$z = \begin{bmatrix} z_2 \\ \vdots \\ z_n \end{bmatrix}$$

$$p_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$



$$\Sigma = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$z \sim \mathcal{N}(0, I) \quad p_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$



En $z = 0$

$$p_z(0) = \frac{1}{\sqrt{2\pi}} e^{-\frac{0^2}{2}} = 0.399$$

En $z = 2$

$$p_z(2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{2^2}{2}} = 0.054$$

Esto es densidad, probabilidad.

$$P(-0.1 < Z < 0.1) \approx 7.97\% \quad \text{vs.} \quad P(1.9 < Z < 2.1) \approx 1.08\%$$

$$P(a < b) = \int_a^b p_z(z) dz$$

Muestra 1

$$z_1 = [0.2, -0.7, \dots]$$

Muestra 2

$$z_2 = [-1.1, 0.3, \dots]$$

Muestra 3

$$z_3 = [0.05, 0.8, \dots]$$

Modelos -GAN

$$G_{\theta}(z) = \hat{x}$$

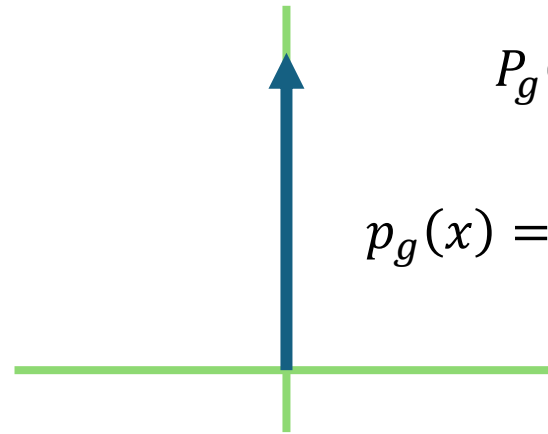
$P_g(x|z)$ **Distribución de probabilidad condicionada**, probabilidad de tener x dado un z

$$p_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

El generador es determinista; es decir, para un mismo vector latente z_0 y un mismo conjunto de pesos θ , la salida generada siempre será la misma.

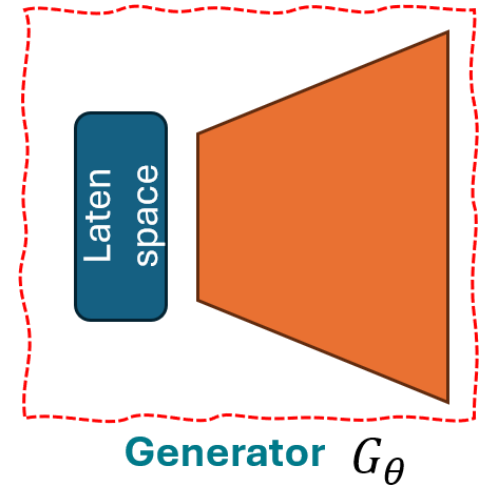
$$p_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$\int_{-\infty}^{\infty} \delta(x - a) dx = 1$$

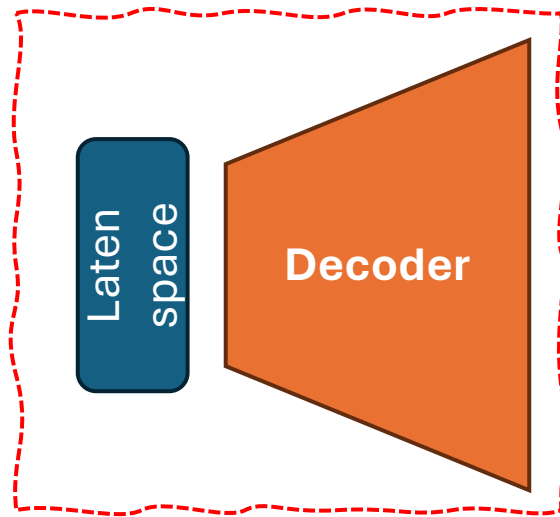


$$P_g(x|z) = \delta(x - G_{\theta}(z))$$

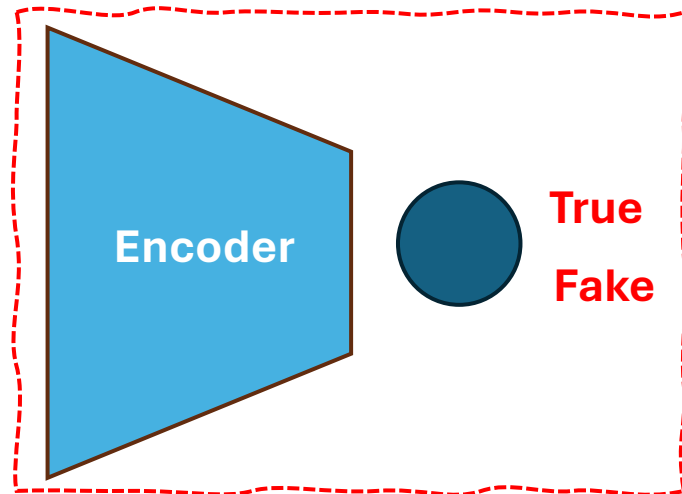
$$p_g(x) = \int_{-\infty}^{\infty} \delta(x - G_{\theta}(z)) p_z(z) dz$$



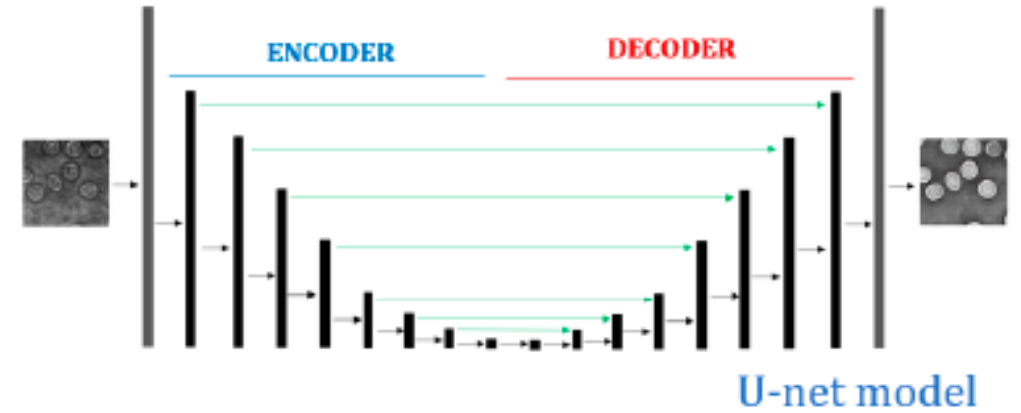
Modelos -GAN

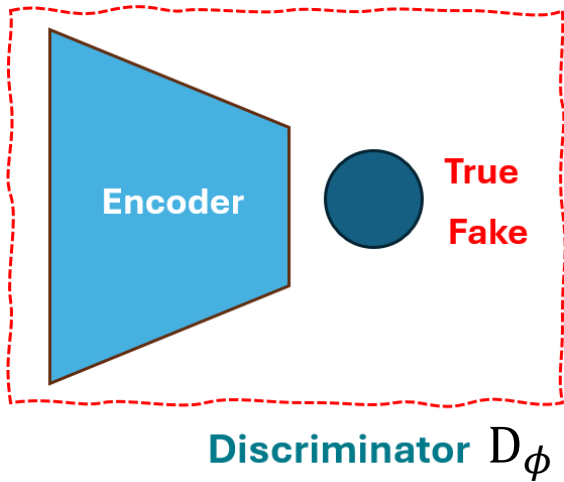


Generator G_θ



Discriminator D_ϕ





$$P(y|x) \longrightarrow D_\phi = P(Y = 1|X = x)$$

$$P(Y = 0|X = x) = 1 - D_\phi$$

Y = 1 | Y = 0

Real Generado

True Fake

Ejemplo: $D_\phi(x) = 0.93$ significa que el modelo asigna 93% de probabilidad a la clase «real».

El discriminador intenta clasificar correctamente ambos tipos de muestra

$$x \sim P_{Data}(x) \longrightarrow D_\phi \longrightarrow Y = 1 \text{ Real}$$

$$G_\theta(z) \longrightarrow \hat{x} \sim P_G(x) \longrightarrow D_\phi(G_\theta(z)) \longrightarrow Y = 0 \text{ Fake}$$

El discriminador convierte “real vs. generado” en un problema de clasificación binaria.

Entropía cruzada binaria (*Binary Cross-Entropy, BCE*)

$$\mathcal{L}(y, \hat{y}) = -[y \text{Log}(\hat{y}) + (1 - y) \text{Log}(1 - \hat{y})]$$

La verosimilitud estadística es una medida de qué tan plausible o creíble es un valor determinado para un parámetro desconocido, basándose en un conjunto de datos ya observado

Modelos -GAN

$$x \sim P_{Data}(x) \longrightarrow D_\phi \longrightarrow Y = 1 \text{ Real}$$

$$G_\theta(z) \longrightarrow \hat{x} \sim P_G(x) \longrightarrow D_\phi(G_\theta(z)) \longrightarrow Y = 0 \text{ Fake}$$

$p_i \rightarrow$ Probabilidad de que el evento ocurra.

$1 - p_i \rightarrow$ Probabilidad de que el evento no ocurra.

$\frac{p_i}{1 - p_i} \rightarrow$ Razón de Probabilidad.

$P(Y = 1 | x) = p$ La probabilidad de que la variable aleatoria Y tome el valor 1, dado el vector de características x , es igual a p .

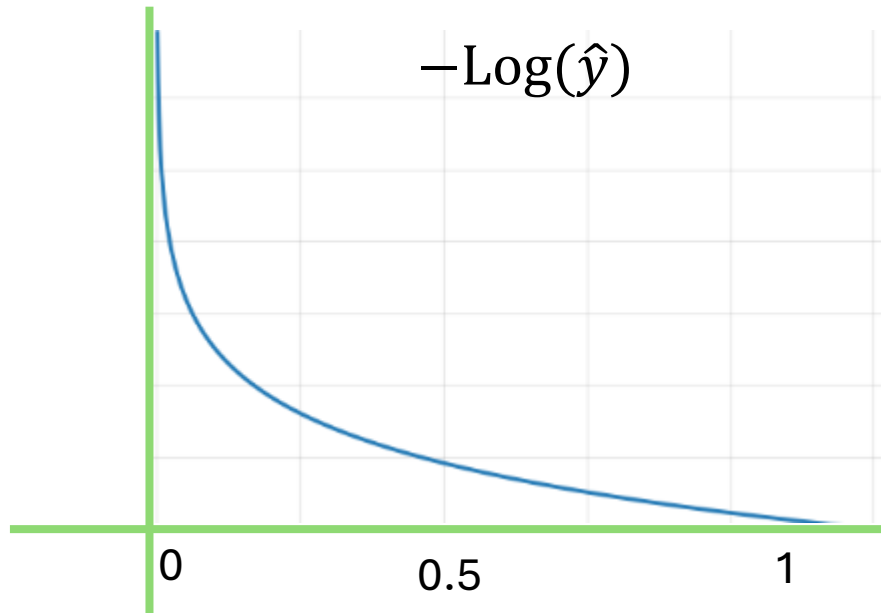
$P(Y = 0 | x) = 1 - p$ La probabilidad de que la variable aleatoria Y tome el valor 0, dado el vector de características x , es igual a $1 - p$.

Bernoulli

$$P(Y = 1 | X = x) = p(x)$$

$$P(Y = 0 | X = x) = 1 - p(x)$$

$$\mathcal{L}(y, \hat{y}) = -[y \text{Log}(\hat{y}) + (1 - y) \text{Log}(1 - \hat{y})]$$



$Y = 1$ **Real**

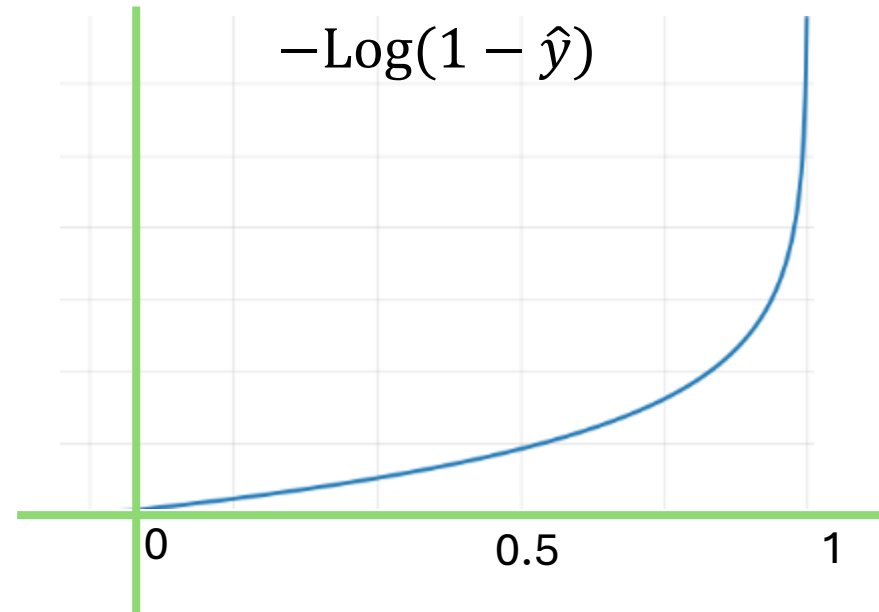
$Y = 0$ **Fake**

Suponiendo imagen real, el discriminador $\rightarrow y = 1$

$$\mathcal{L}(y, \hat{y}) = -[\text{Log}(\hat{y})]$$

$$\hat{y} = 0.9 \rightarrow -\text{Log}(0.9) \approx 0.11$$

$$\hat{y} = 0.1 \rightarrow -\text{Log}(0.1) \approx 2.03$$



Suponiendo imagen falsa, el discriminador $\rightarrow y = 0$

$$\mathcal{L}(y, \hat{y}) = -[\text{Log}(1 - \hat{y})]$$

$$\hat{y} = 0.9 \rightarrow -\text{Log}(1 - 0.9) \approx 2.30$$

$$\hat{y} = 0.1 \rightarrow -\text{Log}(1 - 0.1) \approx 0.10$$

$$\mathcal{L}(y, \hat{y}) = -[y \text{Log}(\hat{y}) + (1 - y) \text{Log}(1 - \hat{y})]$$

$$\mathcal{L}_D = -[\text{Log}(D_\phi(x)) + \text{Log}(1 - D_\phi(G_\theta(z)))]$$

Datos reales

Datos Generado

Si el generador entrega datos falsos que el discriminador toma como reales $D_\phi(G_\theta(z)) = 1$ Se penaliza

Si el generador entrega datos falsos que el discriminador toma como falsos $D_\phi(G_\theta(z)) = 0$ No se penaliza

$$\mathcal{L}_G = -[\text{Log}(D_\phi(G_\theta(z)))]$$

Datos falsos que parece reales $\mathcal{L}_G = -[\text{Log}(1)] = 0$

Datos falsos que no parece reales $\mathcal{L}_G = -[\text{Log}(0)] = \infty$

**BIG
PROBLEM**

Modelos -GAN

$$\mathcal{L}_D = -[\text{Log}(D_\phi(x)) + \text{Log}(1 - D_\phi(G_\theta(z)))]$$

$$\mathcal{L}_G = -[\text{Log}(D_\phi(G_\theta(z)))]$$

A diferencia de una optimización convencional, el entrenamiento de una GAN no busca únicamente el mínimo de una función de pérdida, sino un equilibrio entre dos redes con objetivos adversarios.

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p(x)} [\log D_\phi(x)] + \mathbb{E}_{z \sim p(z)} [\text{Log}(1 - D_\phi(G_\theta(z)))]$$

Datos reales **Datos Generados**

$$\min_{\theta} \max_{\phi} V(\theta, \phi) = \mathbb{E}_{x \sim p(x)} [\log D_\phi(x)] + \mathbb{E}_{z \sim p(z)} [\text{Log}(1 - D_\phi(G_\theta(z)))]$$



PRISIONERO 1

PRISIONERO 2

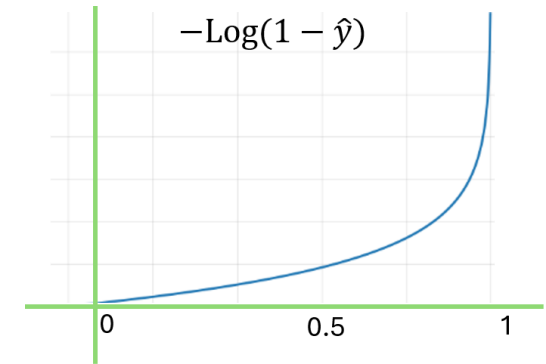
	No delatar	Delatar
No delatar	$(-2, -2)$	$(-10, -1)$
Delatar	$(-1, -10)$	$(-6, -6)$

equilibrio de Nash

El valor esperado es aproximadamente el **promedio sobre el minibatch**:

$$\mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] \approx \frac{1}{m} \sum_{i=1}^m \log D(x_i).$$

$$\mathbb{E}_{z \sim p_z} [\log (1 - D(G(z)))] \approx \frac{1}{m} \sum_{i=1}^m \log (1 - D(G(z_i))).$$

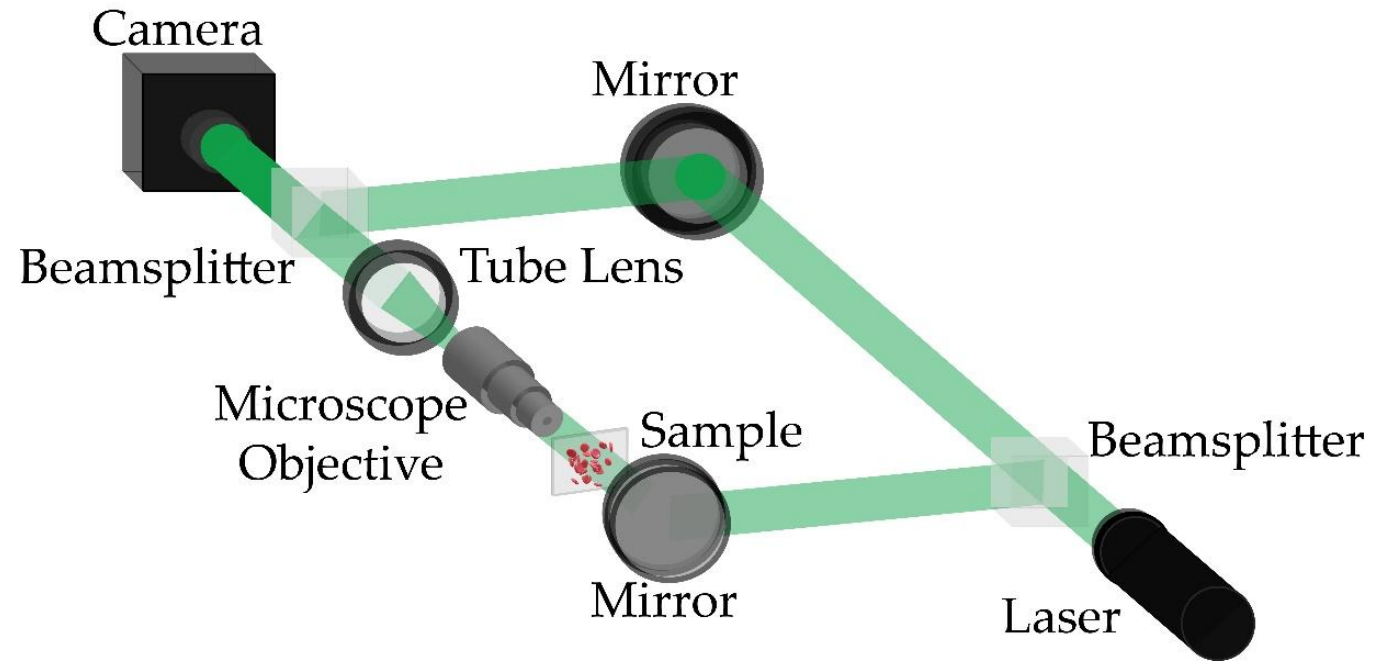
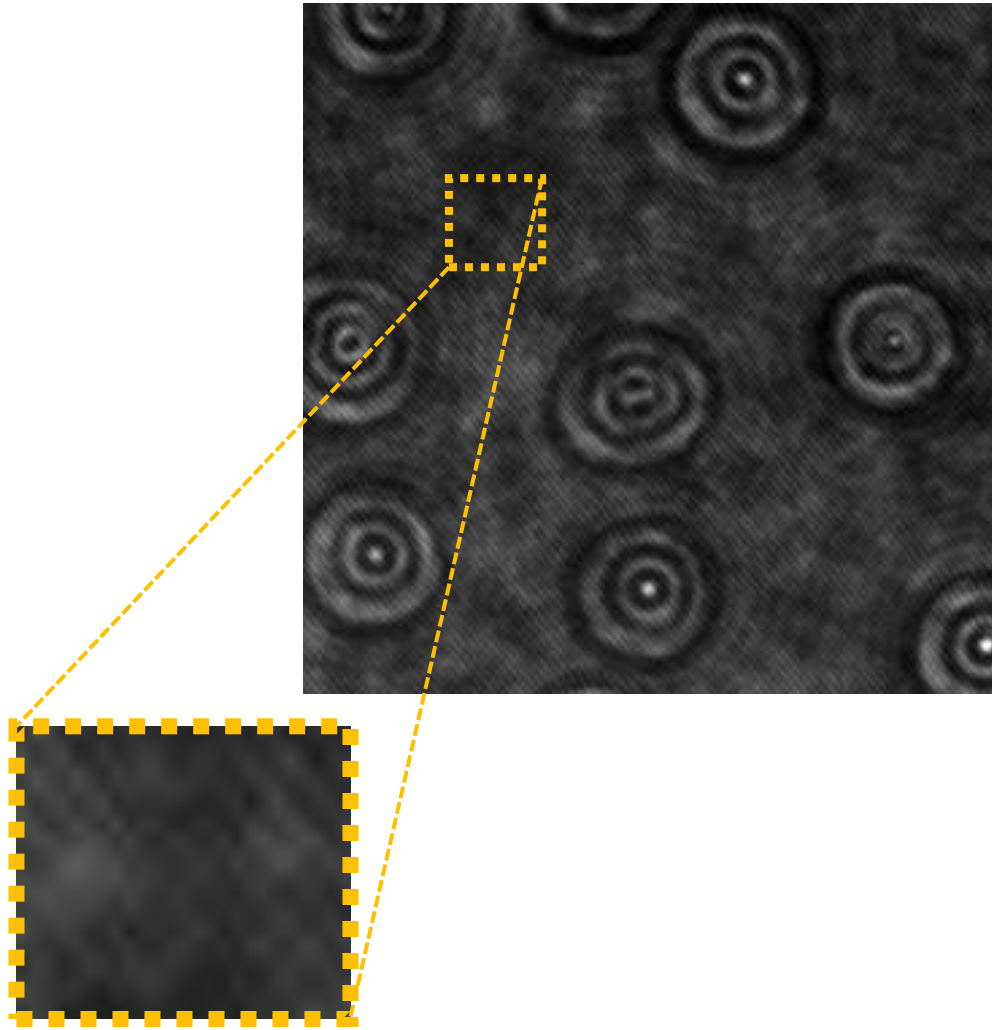


$$\min_{\theta} \max_{\phi} V(\theta, \phi) = \mathbb{E}_{x \sim p(x)} [\log D_{\phi}(x)] + \mathbb{E}_{z \sim p(z)} [\log (1 - D_{\phi}(G_{\theta}(z)))]$$

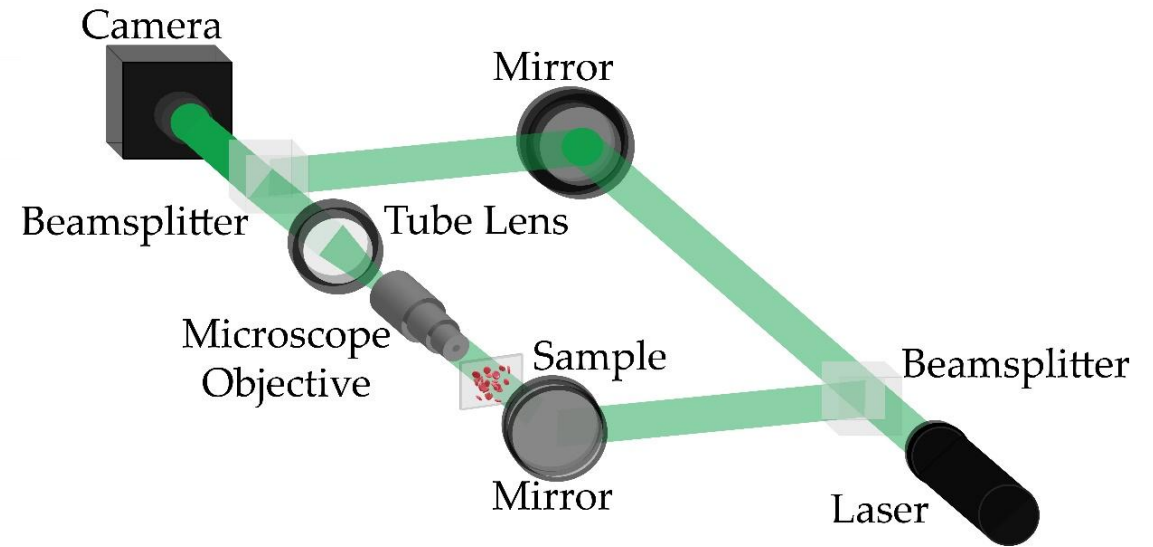
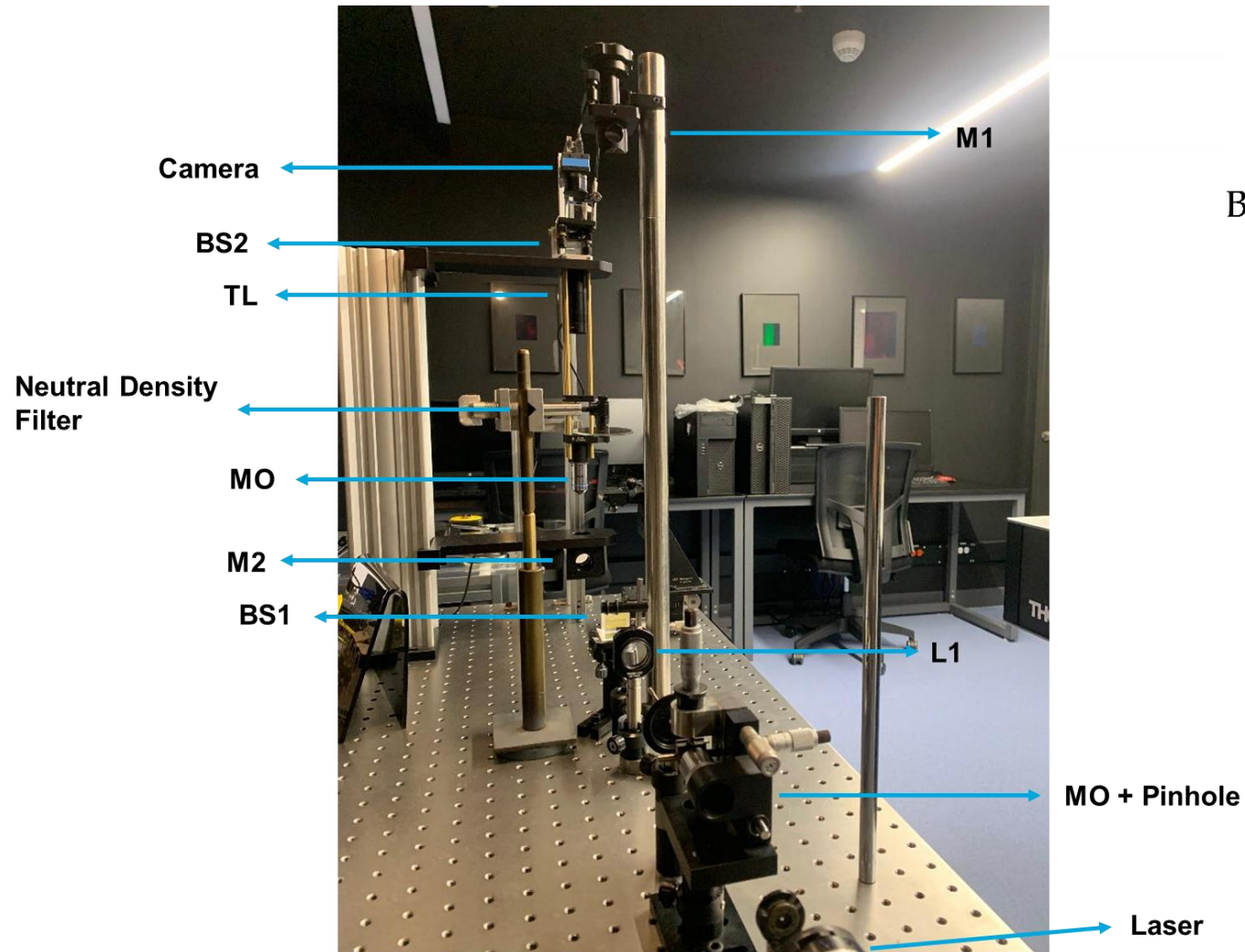
**BIG
PROBLEM**

Al inicio del entrenamiento, el generador produce muestras poco realistas, permitiendo que el discriminador aprenda más rápidamente. Si el discriminador se vuelve demasiado preciso, puede proporcionar gradientes poco informativos al generador, dificultando su aprendizaje.

cGAN Project

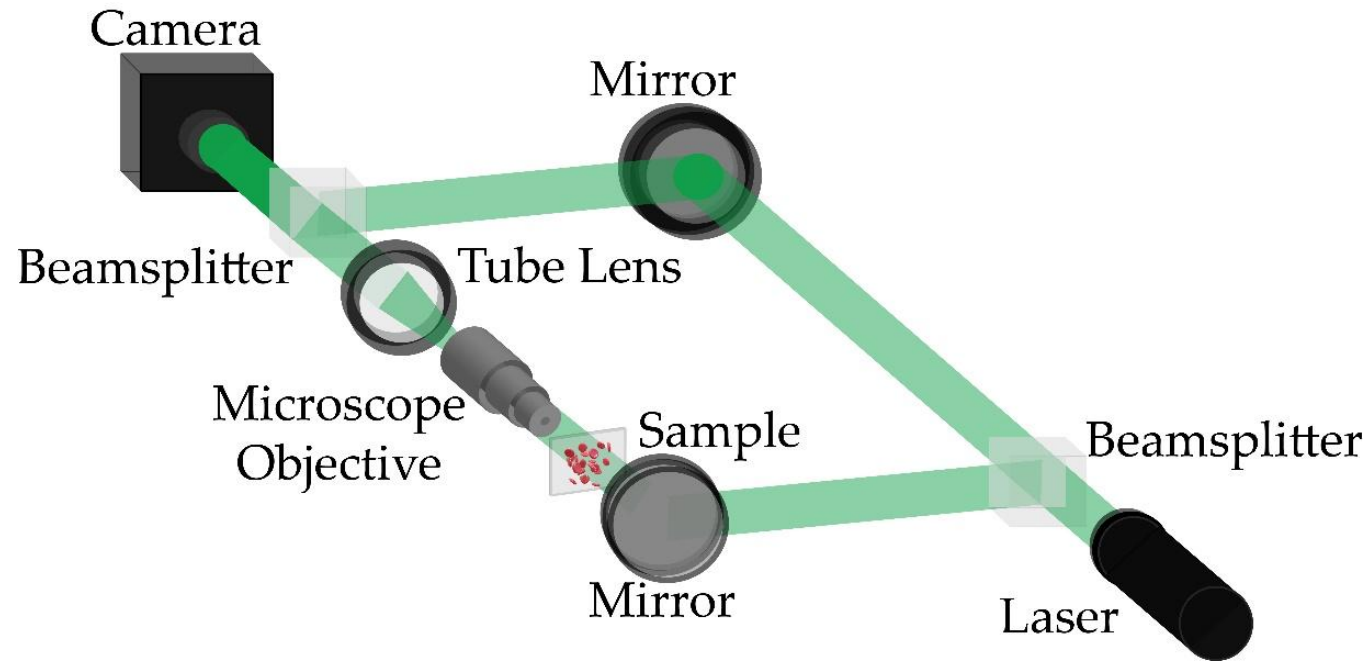


cGAN Project

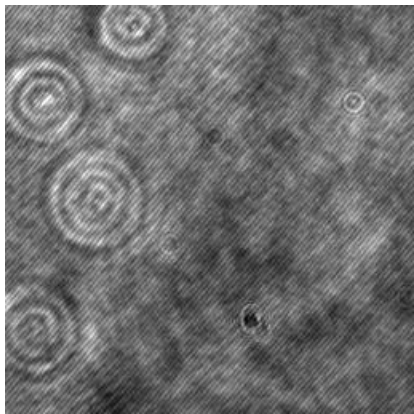


$$h = \frac{\lambda}{2\pi(n_1 - n_2)} \cdot \Delta\phi$$

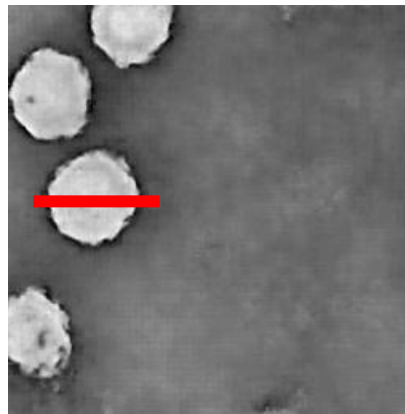
cGAN Project



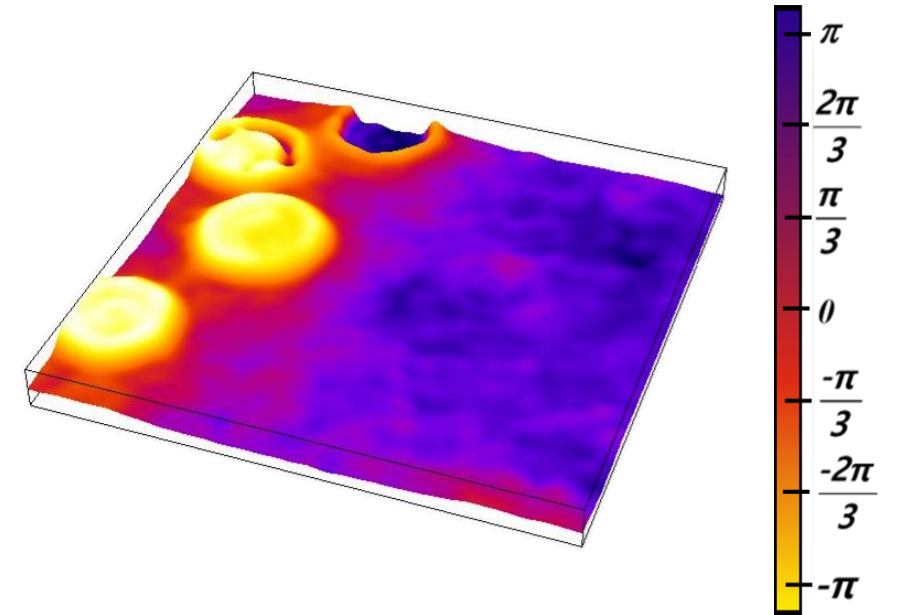
Hologram



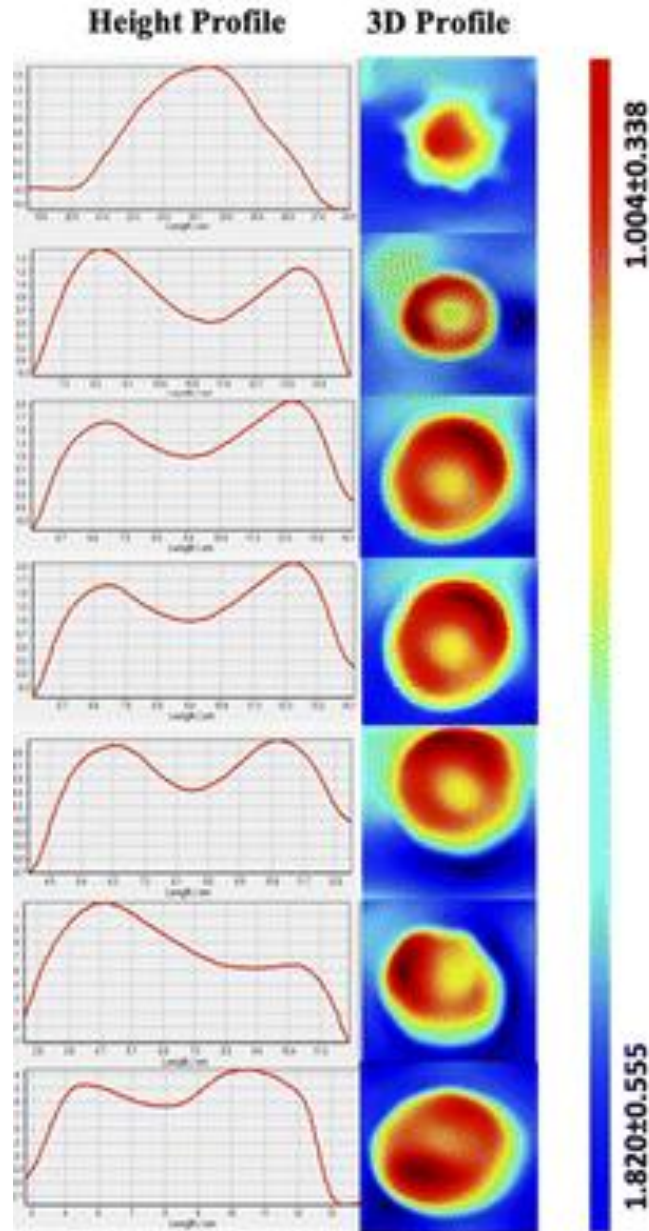
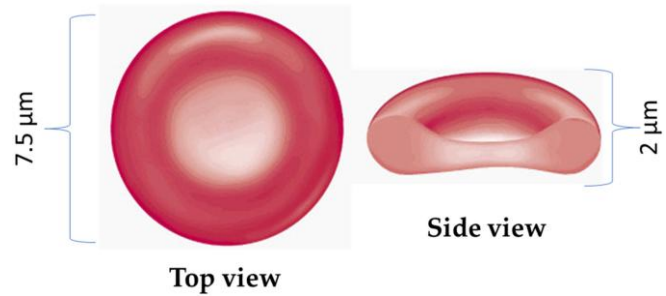
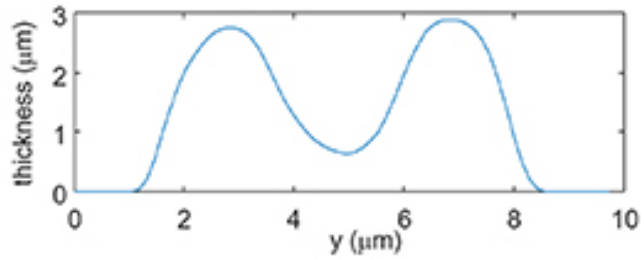
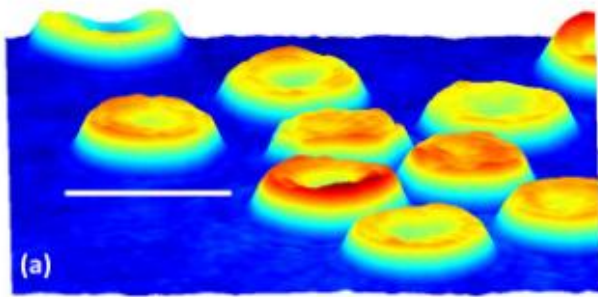
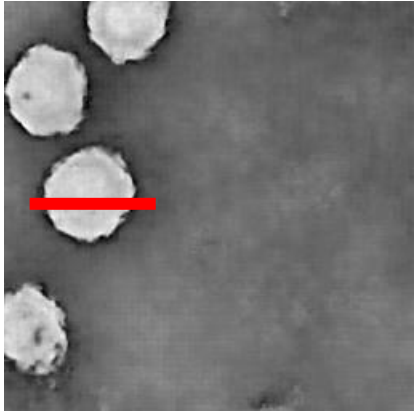
Phase



Numerical
processing

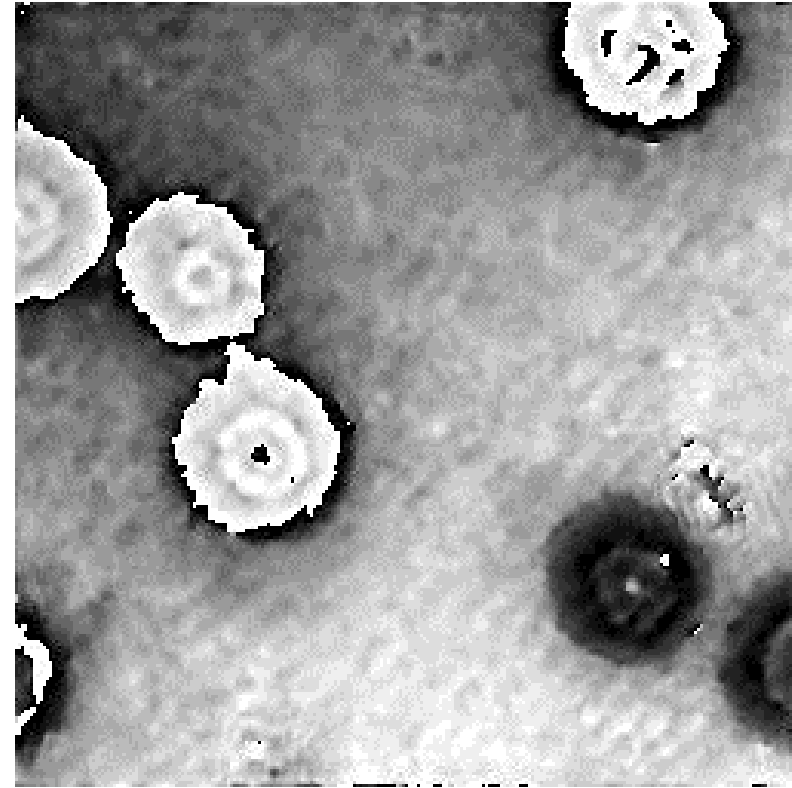
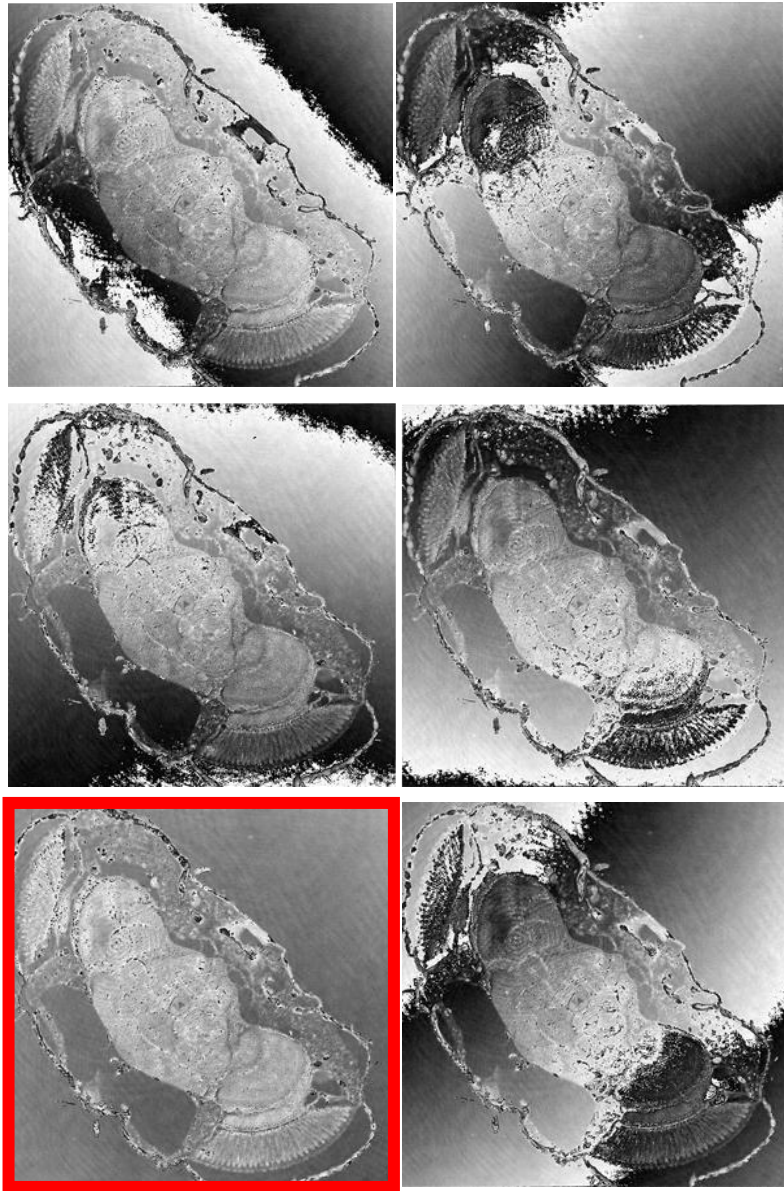


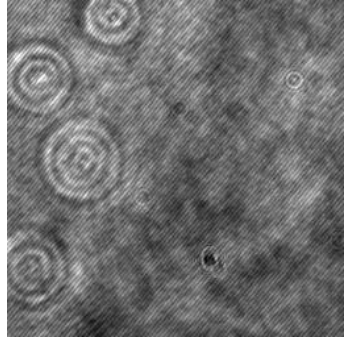
$$h = \frac{\lambda}{2\pi(n_1 - n_2)} \cdot \Delta\phi$$



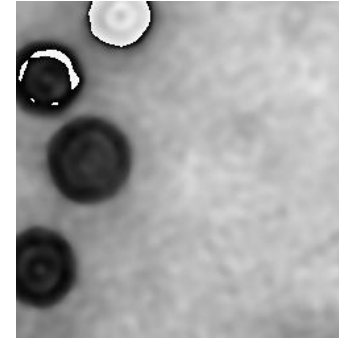
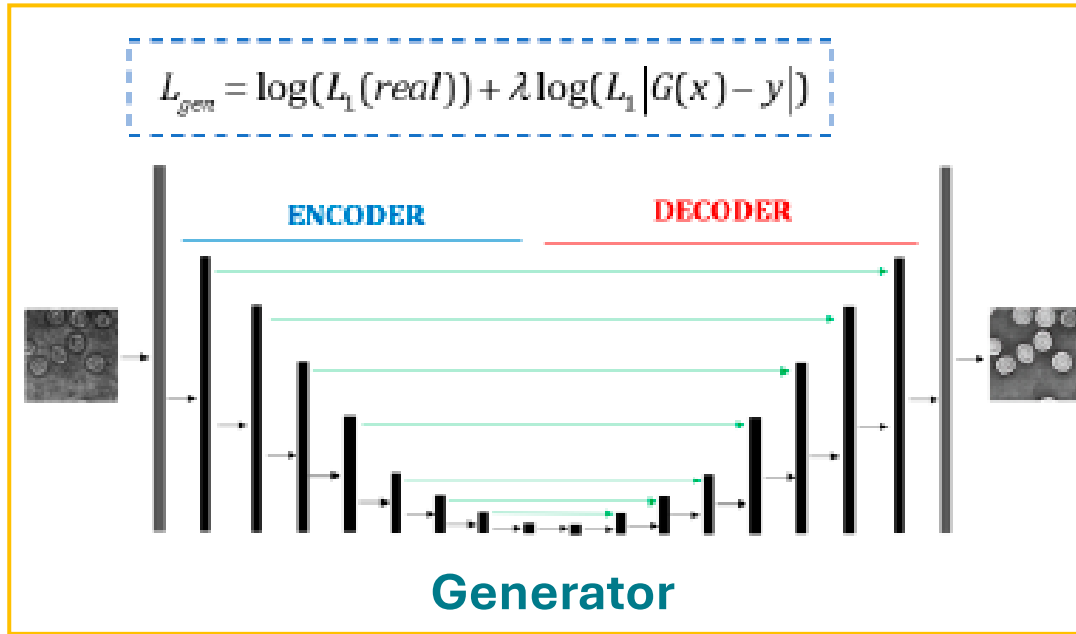
$$h = \frac{\lambda}{2\pi(n_1 - n_2)} \cdot \Delta\phi$$

cGAN Project



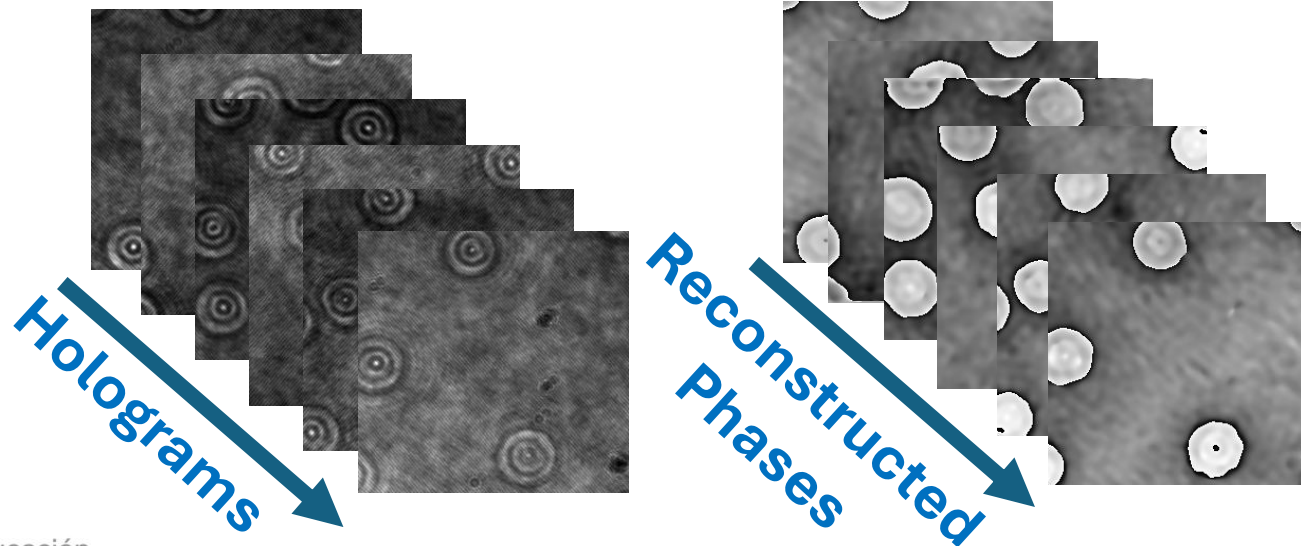


Hologram



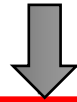
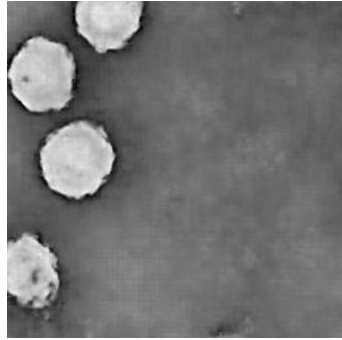
Phase

- ❑ Total number of images: for autoencoders 24,024 instances with images of 256x256 pixels.
- ❑ 80% and 20% of the dataset was used to train and validate the network, respectively.



cGAN Project

Phase
U-net



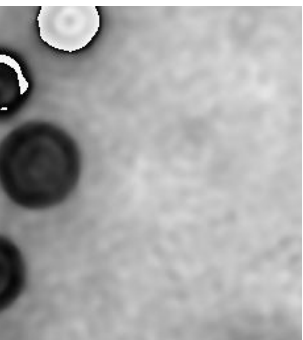
$$L_{real} = \log(L_1(real))$$
$$L_{fake} = \log(L_1(fake))$$

Binary
Classification matrix

Deep CNN
model

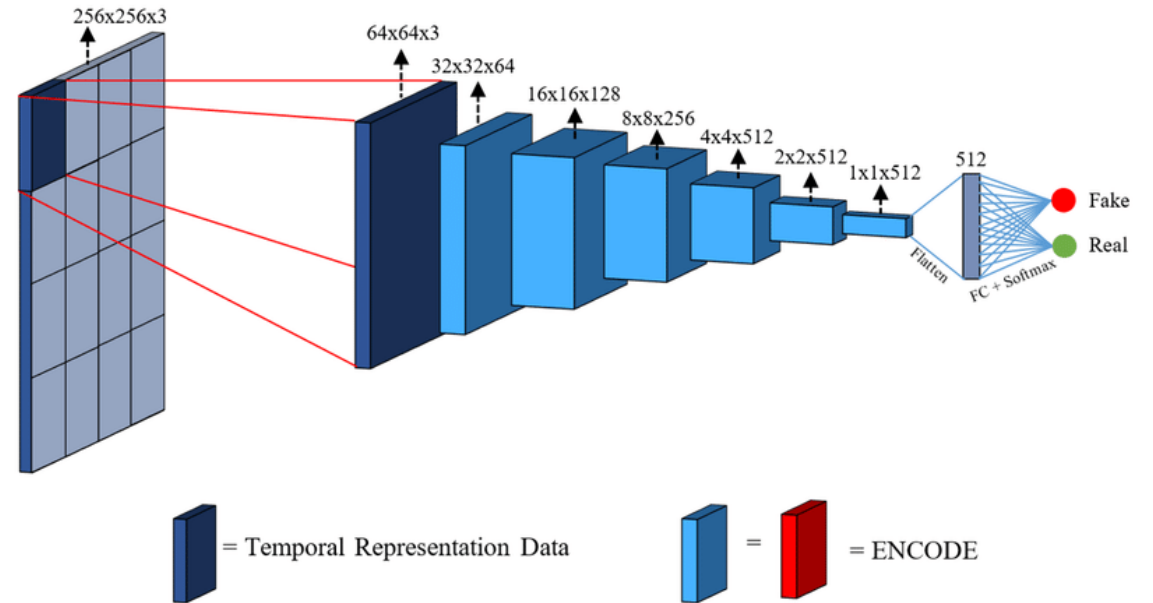
Discriminator

Hologram



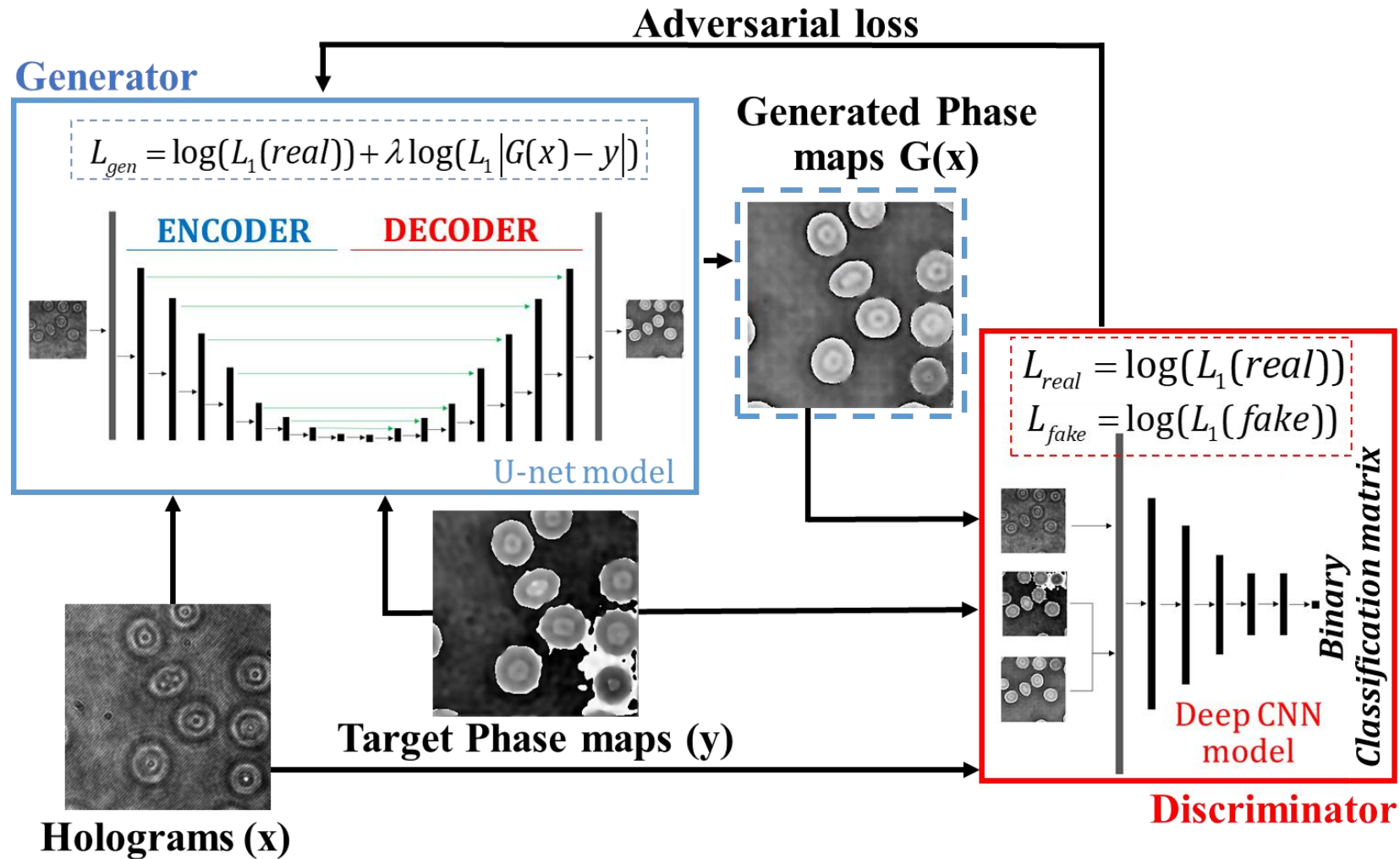
Phase-ground
truth

PatchGAN



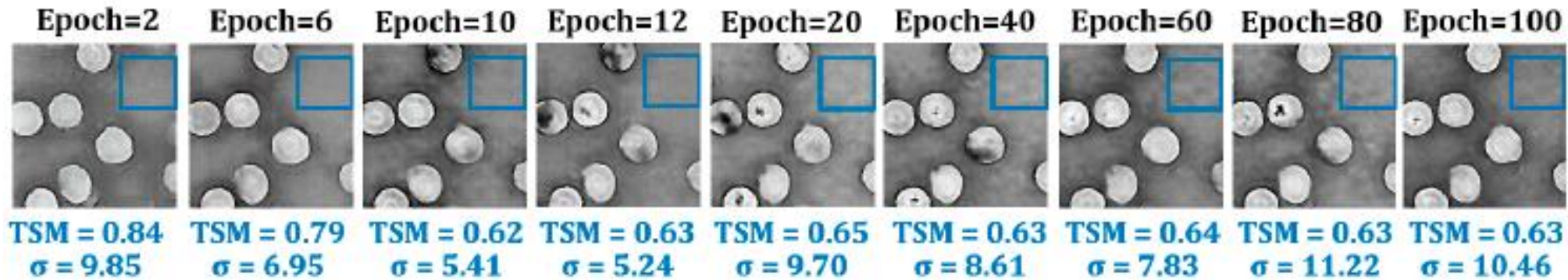
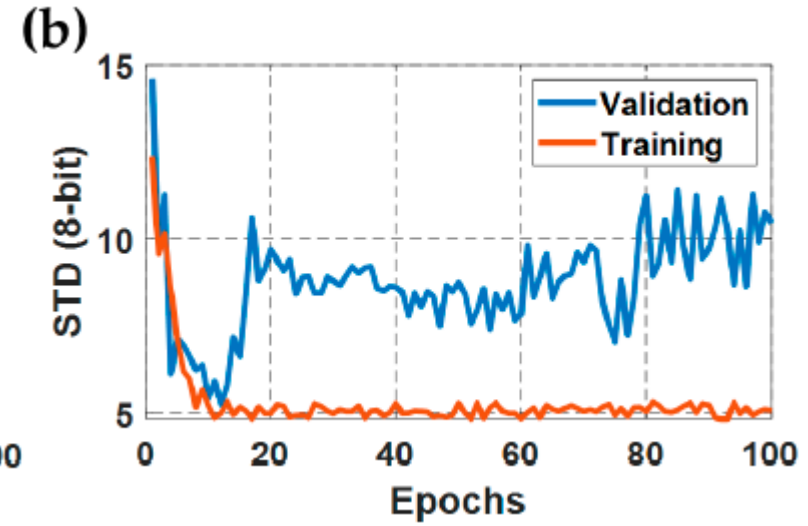
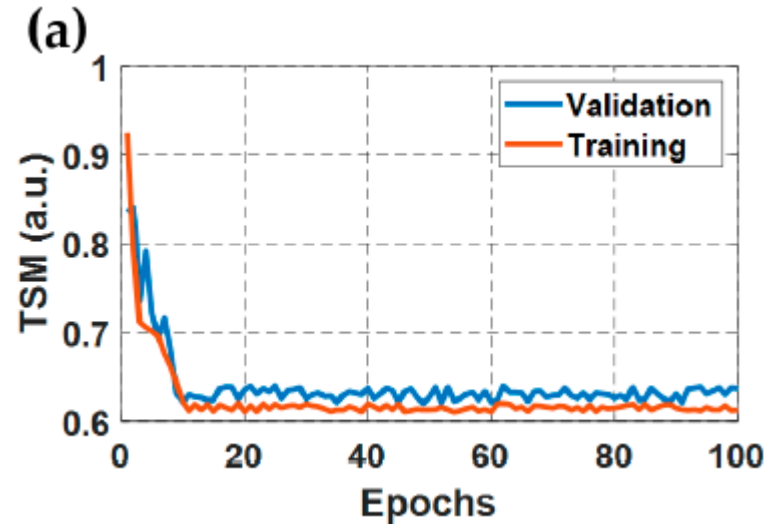
This part is real... this one isn't... this one is...

cGAN Project

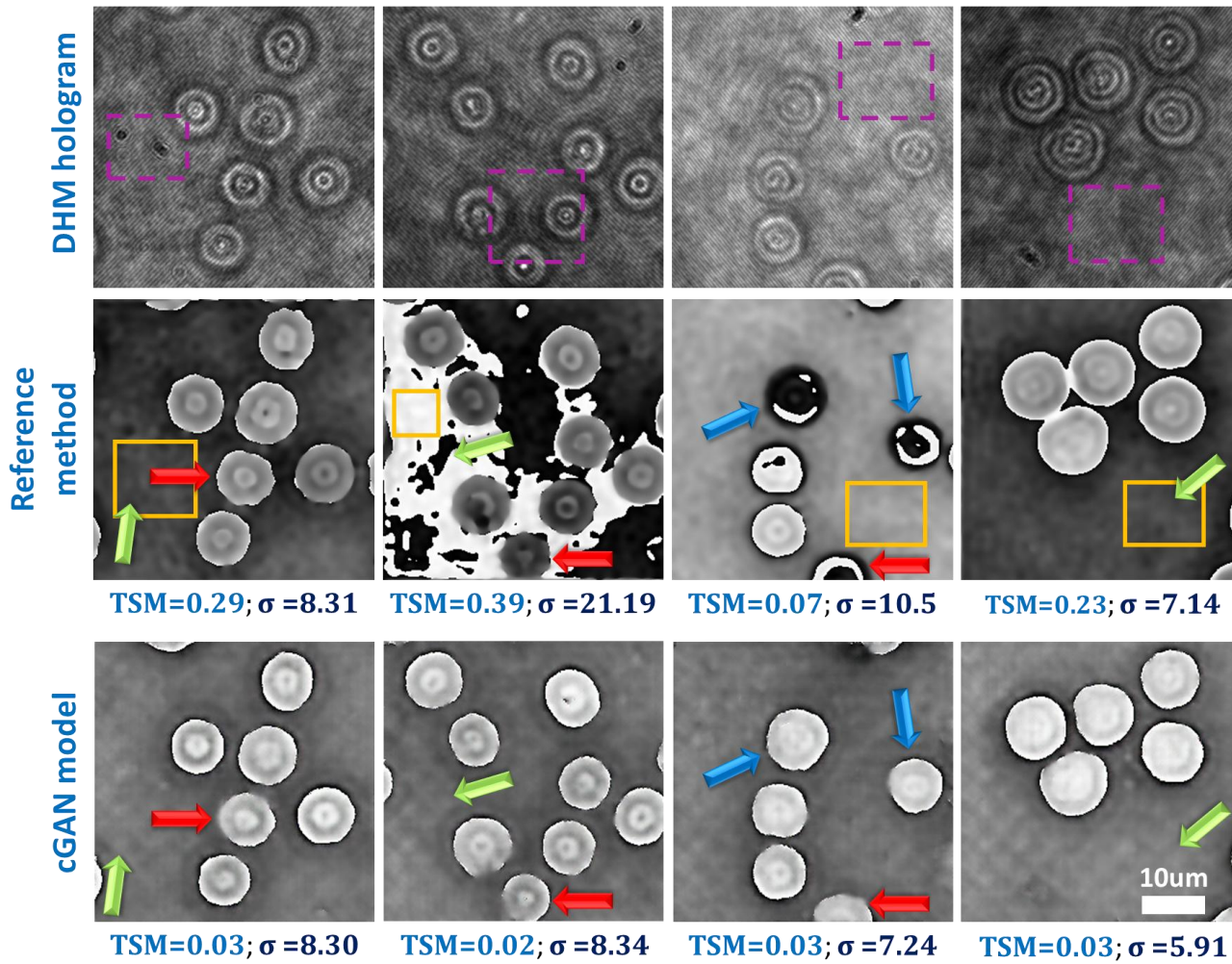


cGAN Project

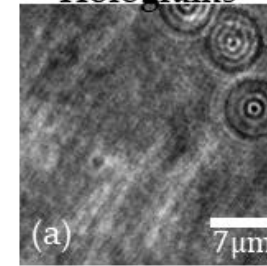
$$\text{TSM}[\hat{h}_{nor}(n, m)] = 1 - \frac{1}{N \cdot M} \sum_{n=0}^N \sum_{m=0}^M \text{Thres}[\hat{h}_{nor}(n, m), 0.2].$$



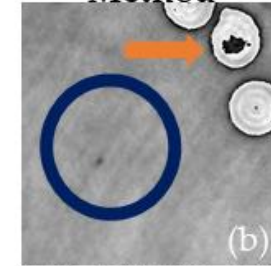
cGAN Project



Fresnel-based DHM Holograms

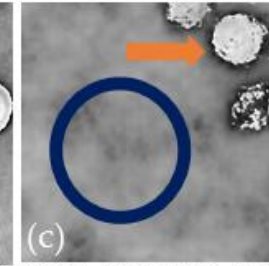


Reference Method



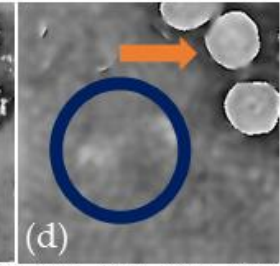
TSM=0.19; STD=8.03

U-Net

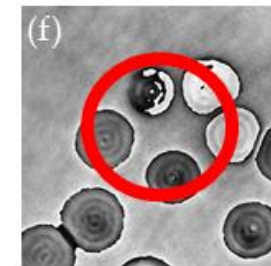
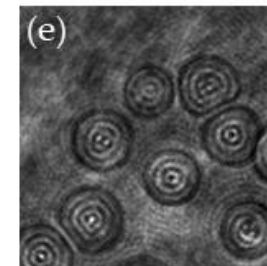


TSM=0.22; STD=7.18

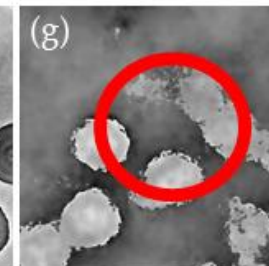
Our cGAN



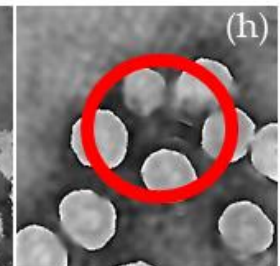
TSM=0.09; STD=7.44



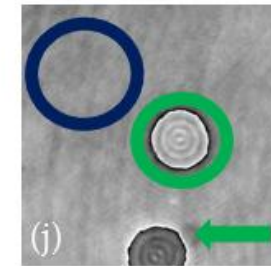
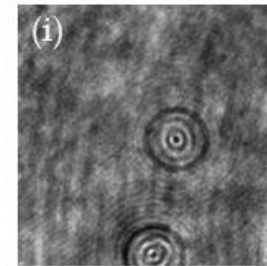
TSM=0.64



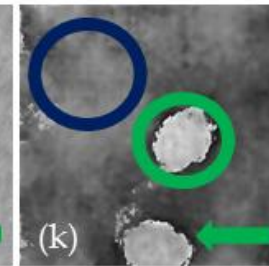
TSM=0.31



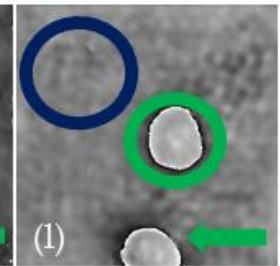
TSM=0.08



TSM=0.37; STD=6.37

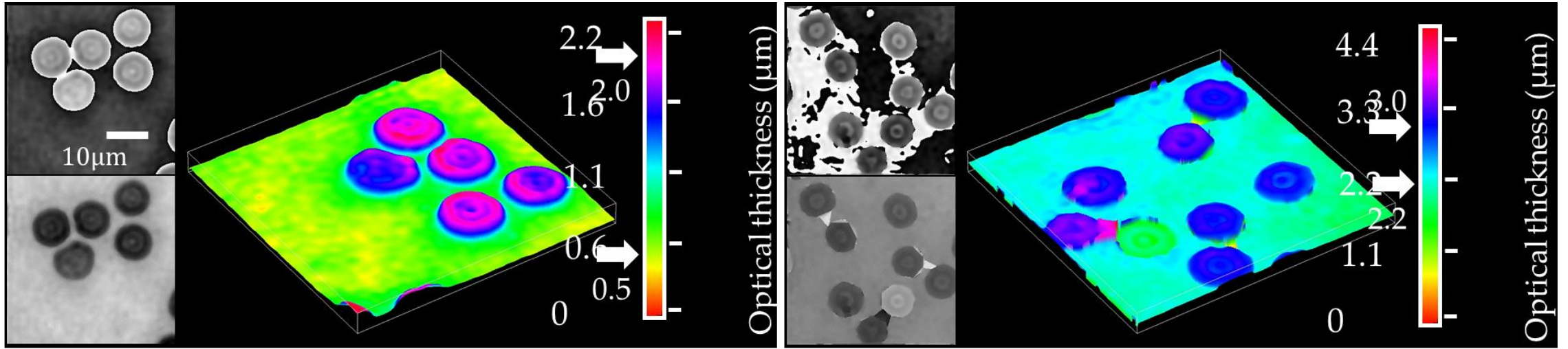


TSM=0.19; STD=7.95

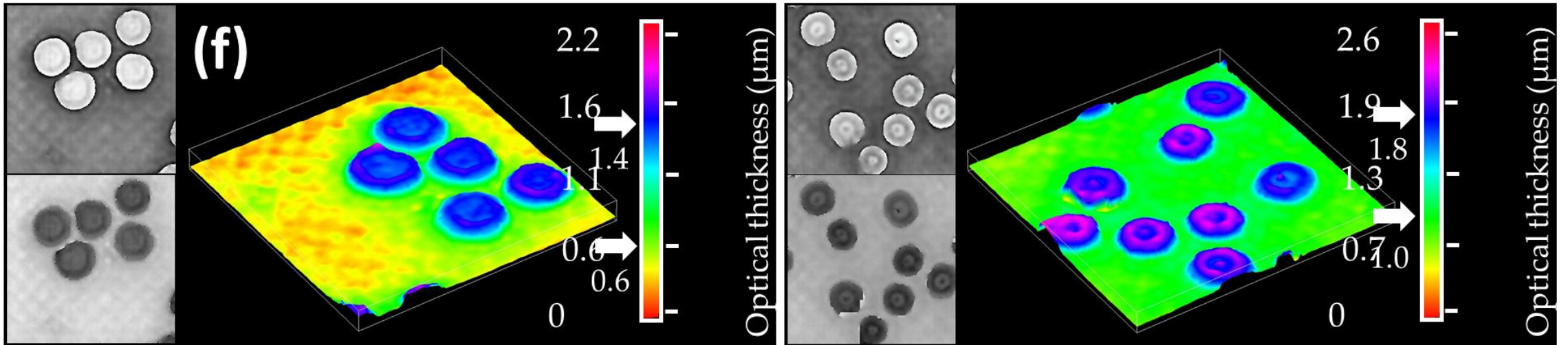


TSM=0.04; STD=9.02

Reference
method



cGAN model



cGAN Project

